

The 66th ASH Annual Meeting Abstracts

POSTER ABSTRACTS

721.ALLOGENEIC TRANSPLANTATION: CONDITIONING REGIMENS, ENGRAFTMENT, AND ACUTE TOXICITIES

Disparities in Microbiota Diversity Impacting Non-European Recipients of 8/8 HLA-Matched Unrelated Donor Transplantation

Natalie Grier Smith, BA^{1,2}, Warren B. Fingrut, MD^{2,3}, Nicholas Waters, PhD⁴, Katarina Piasevoli¹, Teng Fei, PhD⁵, John Slingerland¹, Jonathan U. Peled, MD PhD^{6,2}, Doris M. Ponce, MDMS^{2,7,6}

¹Sloan Kettering Institute, Memorial Sloan Kettering Cancer Center, New York, NY

²Equal Contribution, New York, NY

³Department of Stem Cell Transplantation and Cellular Therapy, The University of Texas MD Anderson Cancer Center, Houston, TX

⁴Department of Immunology, Memorial Sloan Kettering Cancer Center, New York, NY

⁵Department of Epidemiology and Biostatistics, Memorial Sloan Kettering Cancer Center, New York, NY

⁶Adult Bone Marrow Transplantation Service, Memorial Sloan Kettering Cancer Center, New York, NY

⁷Weill Cornell Medical College, New York, NY

Introduction. Although microbiome diversity, patient ancestry, and socioeconomic status (SES) are each associated with allogeneic hematopoietic cell transplantation (allo-HCT) outcomes, the relationships between these variables in the alloHCT context has not been described. We hypothesized that patient ancestry and SES associates with microbiome patterns post-HCT.

Methods. We evaluated alloHCT recipients at our Center 1/2015-6/2020 who provided ≥ 1 stool sample from D-40 to D+365. Patient SES was classified as low if they had an area deprivation index $\geq 60\%$ or were on medicaid or patient financial assistance. Microbial diversity was examined at 4 time points: pre-HCT (D-40 to D-10), engraftment (D+10 to D+19), acute GVHD onset (aGVHD, +/- 7 days), and chronic GVHD landmark (cGVHD, D+100 to D+365). To examine microbial diversity, stool samples underwent 16S V4-V5 region amplification and sequencing, with diversity computed using Simpson's reciprocal (S) for the most representative samples per patient in each timepoint. Butyrate producing anaerobes and strict vs facultative anaerobe ratios were measured. General estimating equations (GEE) modeling was used to investigate interactions between donor type with ancestry and SES. A FLORAL model was used to estimate taxonomic level markers and compare log ratios with a Wilcoxon test.

Results. Overall, 535 patients providing 6017 stool samples were included (median age 61 years, range 50-67; 51% with acute leukemia). 377 (70%) had European and 158 (30%) non-European ancestry (51 African, 40 non-Black Hispanic, 38 Asian, 5 Middle Eastern, 24 mixed non-European). 110 received HLA-matched sibling, 232 8/8 matched unrelated (URD) and 193 HLA-disparate grafts (81 cord blood, 78 haploidentical, 34 $\leq 7/8$ unrelated donors). 151 (28%) received myeloablative and 384 (72%) non-myeloablative/ reduced-intensity HCT. Compared with Europeans, non-Europeans were younger (median 58 vs 65 years, $p=.019$) and received fewer 8/8 URD grafts (33/158, 21% vs 199/377, 53% $p<.001$).

≥ 1 stool sample was provided by 54 patients (32 European, 22 non-European) pre-HCT and by 371 (259 European, 112 non-European) at engraftment, 138 (89 European, 49 non-European) at aGVHD, and 206 (138 European, 68 non-European) at cGVHD timepoints.

Pre-HCT, there were no differences in microbiome diversity by patient ancestry.

Using GEE modeling, considering all timepoints, compared with Europeans, non-European 8/8 URD recipients had lower alpha-diversity ($p=.007$), whereas alpha-diversity was similar by ancestry for all other donor types.

Next we analyzed microbial diversity in 8/8 URD recipients by timepoint. At engraftment, compared with Europeans, non-Europeans had lower alpha diversity ($p = 0.0432$) and lower strict vs facultative anaerobe ratio ($p < .001$) but no difference in butyrate producing anaerobes. When observing taxonomic markers at engraftment by ancestry, non-Europeans had a lower ratio of favorable microbial bacteria including *Lactobacillus/Scardovia* ($p = .005$) and *Akkermansia* ($p = 0.0018$), and a higher abundance of the unfavorable *Escherichia/Phascolarctobacterium* ($p = 0.0023$). At aGVHD and cGVHD timepoints, there were no differences in alpha diversity by patient ancestry. However, at aGVHD, compared with Europeans, non-Europeans had lower

strict versus facultative anaerobe ratio ($p = 0.0156$). Non-Europeans also had decreased butyrate producing anaerobes at both the aGVHD ($p < .001$) and cGVHD ($p = .0175$) timepoints.

There were no differences in microbiome composition between groups by SES at any time point.

Conclusions. Our analyses identified disparities in microbiome diversity across multiple timepoints post-HCT which impact non-European ancestry 8/8 URD recipients. Prospective research is needed to investigate potential causes for these differences, considering antibiotic exposure, diet/lifestyle factors, and more detailed SES evaluation.

Disclosures Peled: *Prodigy Biosciences*: Current equity holder in publicly-traded company, Membership on an entity's Board of Directors or advisory committees; *Canaccord Genuity, Inc*: Consultancy; *Postbiotics Plus Research*: Current equity holder in publicly-traded company, Membership on an entity's Board of Directors or advisory committees; *MaaT Pharma*: Consultancy; *CSL Behring*: Consultancy; *Crestone Inc*: Consultancy; *Seres Therapeutics*: Patents & Royalties, Research Funding; *DaVolterra*: Consultancy. **Ponce:** *Incyte*: Consultancy, Membership on an entity's Board of Directors or advisory committees, Research Funding; *Sanofi*: Consultancy, Membership on an entity's Board of Directors or advisory committees, Research Funding; *Evive*: Consultancy; *OncLive*: Consultancy.

<https://doi.org/10.1182/blood-2024-212186>